NPP INSTITUTE

Student Missed Class Recovery & Academic Attendance Planner

Generated: Fall 2026 Term

1. EXECUTIVE SUMMARY & OBJECTIVES

The NPP Institute Missed Class Recovery Planner is a collegiate academic application designed to eliminate student course debt resulting from unavoidable lecture or laboratory absences. It converts sudden missed class periods into structured, time-blocked 72-hour recovery dossiers, coordinates peer note sharing, automates professional communication with instructors, and tracks mandatory institutional attendance thresholds to prevent exam debarment.

2. CORE APPLICATION FEATURES

A. Automated Absence Triage & Recovery Dossiers

Generates customized milestone checklists depending on class type (Lecture, Lab practical, Seminar, or Recitation). Automatically factors in slide decks, Panopto recording replays, and problem sets.

B. Mandatory Attendance Risk & Debarment Calculator

Tracks institutional attendance compliance (75% and 80% thresholds). Features an interactive "What-If" trajectory simulator calculating the exact number of consecutive classes a student must attend to restore compliant standing.

C. Study Sprint Focus Mode (Pomodoro Timer)

Integrated 15m, 25m, and 45m deep study timers linked directly to recovery tasks. Includes soft Web Audio completion chimes and auto-saving scratchpads for instructor office hour doubts.

D. Professor & TA Communication Suite

One-click email generator supporting 5 academic scenarios: Medical clinic notice, quiz/lab makeup inquiry, conference conflict, personal emergency, and office hours review. Offers Formal and Concise tone presets.

E. Peer Notes Network & Reciprocal Academic Karma

Directory of trusted classmates for note sharing, status tracking (requested, received, reviewed), and one-click SMS request generation.

F. University Single Sign-On (SSO) & Demo Personas

Secure authentication via Google Workspace for Education, .edu credentials, and instant 1-click student personas (Alex Mercer - CS, Jordan Taylor - Pre-Med, Taylor Reed - Economics).

3. ENROLLED COURSE AUDIT & ATTENDANCE METRICS

Course Code & Title	Schedule	Attended/Held	Rate	Status
CS 201: Data Structures	MWF 10:00-11:15 AM	14 / 16 held	87.5%	Safe
BIO 110: Cell Biology & Gen	TTh 01:30-03:00 PM	9 / 12 held	75.0%	**Danger (<80%)**
MATH 215: Linear Algebra	TTh 10:00-11:30 AM	10 / 11 held	90.9%	Safe
ECON 102: Macroeconomics	MWF 02:00-03:15 PM	15 / 15 held	100.0%	Safe

4. STANDARD 72-HOUR RECOVERY PROTOCOL

0 – 12 Hours Post-Absence

Transmit polite absence notice to instructor or TA. Connect with registered peer note partner to request lecture annotations.

12 – 36 Hours Post-Absence

Stream recorded lecture at 1.25x or review slide deck. Cross-examine peer notes and solve 2 foundational practice problems.

36 – 72 Hours (Prior to Next Class)

Complete homework or lab analysis. Test key concepts with the self-check mastery tool. Formulate 2 questions for upcoming office hours.

5. ARCHITECTURE, SECURITY & FERPA COMPLIANCE

• Client Framework: React 19 SPA running on Vite with TypeScript and Tailwind CSS.
• Security & Auth: Firebase Authentication (Google OAuth + Email/Password) with FERPA-aligned access policies.
• Data Persistence: Google Cloud Firestore with Attribute-Based Access Control (ABAC) rules isolating student records.
• Offline Resilience: Hybrid localStorage synchronization ensuring offline task progress retention.
• Audio & Interactivity: Native Web Audio API synthesis for distraction-free sprint cues.